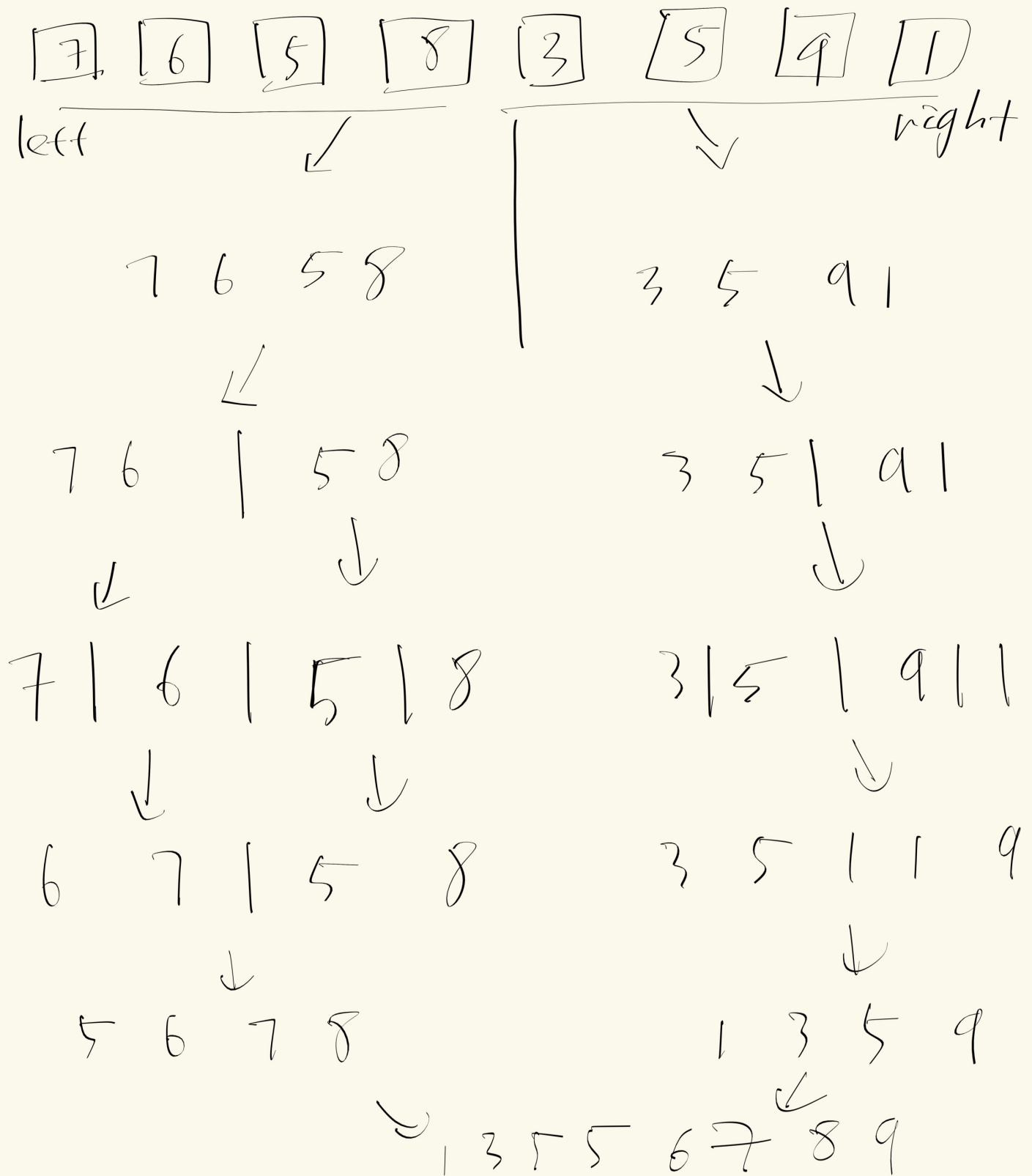


Merge Sort

- $O(n \log n)$
- Divide and Conquer



```
def merge_sort(arr):
```

```
    if len(arr) == 1:
```

```
        return arr
```

```
    mid = len(arr) // 2
```

```
    left_arr = merge_sort(arr[:mid])
```

```
    right_arr = merge_sort(arr[mid:])
```

```
    return merge(left_arr, right_arr)
```

```
def merge(left_arr, right_arr):
```

```
    res = []
```

```
    i = j = 0
```

```
    while i < len(left_arr) and j < len(right_arr):
```

```
        if left_arr[i] < right_arr[j]:
```

```
            res.append(left_arr[i])
```

```
            i += 1
```

```
        else:
```

```
            res.append(right_arr[j])
```

```
            j += 1
```

```
    res.extend(left_arr[i:])
```

```
    res.extend(right_arr[j:])
```

```
    return res
```